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Bisphenol F promotes the secretion of pro-inflammatory cytokines in macrophages by enhanced glycolysis through PI3K-AKT signaling pathway

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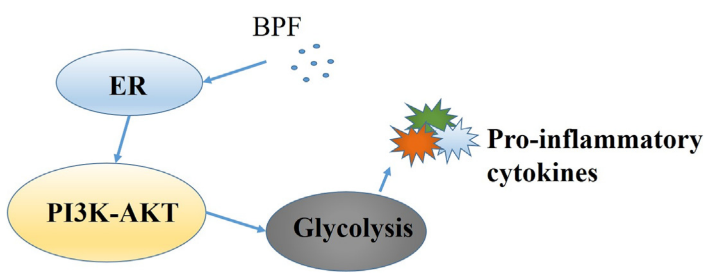
HIGHLIGHTS

- BPF promotes glycolysis and the secretion of pro-inflammatory cytokines in macrophages.
- Enhanced glycolysis promotes the secretion of pro-inflammatory cytokines induced by BPF.
- The PI3K-AKT pathway is involved in BPF induced glycolysis and the secretion of pro-inflammatory cytokines of macrophages.
- Estrogen receptor (ER) mediates the activation of PI3K-AKT signaling pathway induced by BPF.

GRAPHICAL ABSTRACT



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ABSTRACT

Bisphenol F (BPF) is a member of endocrine disrupting chemicals (EDCs). As a substitute of bisphenol A (BPA), BPF is widely used in various consumer products, leading to an increased risk of people's exposure. However, there are few studies on the immunotoxicity and mechanism of BPF. This study aimed to investigate the effect of BPF on the secretion of pro-inflammatory cytokines by macrophages and explore its mechanism. In our study, RAW264.7 macrophages were treated with different concentrations of BPF (0, 5, 10 and 20 μM) for 24 h. The results showed that the secretion of pro-inflammatory cytokines (IL-6, TNF-α and IL-1β) and the production of lactate were increased in a dose-dependent manner. BPFalso led to the activation of the PI3K-AKT signaling pathway. After pretreatment with glycolysis inhibitor (2-DG) and exposure to BPF (20 μM), the secretion of pro-inflammatory cytokines induced by BPF was inhibited. PI3K inhibitor (LY294002) and estrogen receptor (ER) antagonist (ICI 182,780) could also inhibit the above effects induced by BPF (20 μM). In conclusion, our results suggested that BPF can enhance

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